

Katherine Avanesov

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Education

California Institute of Technology (Pasadena, CA September 2023 - June 2027)

- Physics (Major) and Computer Science (Minor).
- Cumulative GPA: 4.0/4.0
- Relevant coursework: Operating Systems, Software Engineering, Database Systems, Machine Learning, Computational Physics, Quantum Computation, Mathematical Chaos, Mathematical Methods in Physics.

Experience

Quantitative Research Intern – Clear Street (Summer 2026).

- Deployed a Python risk engine to production, integrating with Snowflake database via SQL for daily recalibration with a front-end dashboard providing risk managers with per-position cost breakdowns.
- Developed an option volatility-risk model spanning a 6,000+ stock universe to estimate liquidation cost and set margin requirements for institutional portfolios.
- Designed an algorithm that recognizes offsetting risks across a portfolio, reducing over-conservative margin estimates by 20%+ for diversified positions by decomposing implied-volatility fluctuations into normal modes of a multidimensional risk surface.

Researcher – CERN / Princeton CMS Group (Summers '24 – '25).

- 10 weeks on-site at CERN in Geneva, Switzerland under short-term User contract.
- Led the development of neural network-based classifiers for particle detection, improving signal selection by over two orders of magnitude; applied Grad-CAM to interpret 3D Convolutional Neural Network output, and designed statistical framework for background estimation.
- Presented at American Physical Society Global Physics Summit 2025: Search for Long-Lived Particles with Hadronic Calorimeter Depth Segmentation.

Projects

“NanoDB”: SQL Database Engine (Winter 2026).

- Built core components of a relational database across a 75,000+ line Java codebase spanning 15 subsystems, including low-level file storage, B+ tree indexes, and a query execution pipeline.
- Implemented external-memory algorithms, table-statistics tracking, and a dynamic-programming query optimizer that used those statistics to estimate execution costs and select efficient execution plans.

Java Virtual Machine (Fall 2025).

- Implemented a minimal Java Virtual Machine supporting class-file parsing, bytecode execution, and operand stack management. Validated the interpreter on sorting algorithms, recursive programs, and larger workloads including Collatz trajectories for 100k integers.

Spreadsheet Engine (Winter 2026).

- Collaborated with a partner over 10 weeks to build a spreadsheet engine supporting formula evaluation, function invocation, dependency tracking, and circular-reference detection.
- Improved runtime performance by up to 10x by optimizing the underlying cell-dependency graph with an iterative topological sort algorithm and batched cell-update logic.

Automated Anomaly Detection System for Data Quality Monitoring at CERN (Summer 2024).

- Developed a web-based autoencoder anomaly detection system to assist in real-time data quality monitoring for LHC collisions (~40 million per second).
- System partially integrated into the Run Registry, with ongoing testing before deployment in the official Data Quality Monitoring (DQM) workflow.

“Doodle Jump” (Spring 2025).

- Built a C physics engine with collision detection and gravity to create a game similar to the original “Doodle Jump”, using the SDL library for rendering and input, and created an HTTP server to host game online. Features include computer-controlled enemies, progressive difficulty levels, and menu selection managed by decision-tree logic.

Anomaly & Wear Detection with Tool Vibrational Data (Spring 2025).

- Developed a method for detecting CNC tool wear using vibrational waveform data from cyclic machining processes. Detected anomalous tools by evaluating multiple metrics, such as RMS differences between cycles and correlation, to capture both instantaneous breakage events and gradual tool wear; performed PCA on the combined metrics for each tool and K-means clustering in the resulting space to isolate outlier tools.

ML-Enhanced Momentum Strategy (Spring 2025).

- Developed and back-tested a long-short equity momentum strategy in Python, using Pandas and LightGBM Random Forest regressor with features of multiple momentum horizons and firm-level metrics. Performed regressions and risk analysis to evaluate return persistence, robustness across regimes, and liquidity constraints. Achieved statistically significant alpha (~7% annualised) above standard Fama-French momentum strategy.

Teaching, Leadership, and Volunteering Experience

Teaching Assistant – Caltech (Winter - Spring 2026):

- Supported 100+ students in Caltech’s foundational courses CS2 (Introduction to Programming Methods, Java) and CS3 (Introduction to Software Design, C) through weekly office hours, providing conceptual and debugging help.

Student Advisory Board Member – Caltech (Fall 2025-Spring 2026)

- Attended monthly meetings to provide student perspective and help plan programs for Caltech’s PMA department, including career fairs, wellness events, and game nights.
- Served on a panel for prospective Caltech freshmen, speaking about my academic and student experience.

Gymnastics Coach and Translator – Sevenoaks Gymnastics Club (2021-2023)

- Started as a volunteer and later became a paid gymnastics coach at my local club, coaching children aged 3–9, providing physical support, and ensuring their safety during classes.
- Served as an English–Russian translator for an incoming Ukrainian refugee attending classes.

Middle School Physics Club – Sevenoaks School (2021-2023)

- Founded a weekly club for students aged 11–13 to introduce advanced physics concepts in an accessible way, emphasizing both mathematical methods and interactive experiments and demonstrations.
- Designed 30+ original physics problems inspired by fun contexts, such as modelling the “physics” of Disney movies, to engage students with topics ranging from calculus-based dynamics, including simple harmonic motion, to thermodynamics and special relativity.

Skills

- Languages: Python, C, Java, SQL. Fluent Spanish, French, Russian. Elementary Japanese.
- Tools: Git, Linux, Pandas, TensorFlow, PyTorch, Snowflake, Datadog

Awards

- 2026 The George W. and Bernice E. Green Memorial Prize nominee for creative scholarship beyond coursework
- 2025 Caltech Margie Lauritsen Leighton Prize for academic excellence in Physics
- 2025 Caltech Hallet Smith Prize for best essay on Literature
- 2025 Bill Gross Business Plan Competition Finalist
- 2025 Taylor Lawrence SURF Research Fellow
- 2024 Caltech Robert F. Christy Prize for an Outstanding Freshman in Physics
- 2024 Samuel P. and Frances Krown SURF Research Fellow
- 2023 Top Gold Round 1 and Silver Round 2 British Physics Olympiad; Gold British Chemistry Olympiad
- 2022 Gold Senior Physics Challenge (Awarded by British Physics Olympiad Organization)